

Prithvi Raj

🔗 Prithvi-R | 📧 Prithvi Raj | rprithvi939@gmail.com | 📞 +91 9955095089

Education

Indian Institute of Information Technology, Guwahati
B.Tech in Electronics and Communication Engineering

Aug 2023 – Current
CGPA: 7.5

Technical Skills

Languages: Python, C++, JavaScript, Java, Kotlin, C, SQL
AI & ML: Scikit-learn, Keras, Pandas, NumPy, OpenCV, NLP
Web Development: React.js, Node.js, Flask, REST APIs, HTML5, CSS3
Tools & Platforms: Git, GitHub, Google Cloud (Gemini API)

Research & Core Projects

Project Ascend: AI-Powered Life Tracker [Github] [Live Demo] (Personal Project) July 2025

- Programmed a full-stack wellness tracker using FastAPI and React, gamifying daily habits with AI-generated side-quests.
- Logged activities (nutrition, exercise, mood) and generated weekly reports via a modular backend with JWT auth and PostgreSQL.

Terminal Based Portfolio [Github] [Live Demo] (Personal Project) July 2025

- Orchestrated a terminal-themed portfolio featuring an interactive CLI where users can query my qualifications using custom commands.
- Built an intelligent Q&A system with a custom response-ranking algorithm to provide detailed information.

Portable Spectrophotometer for Turmeric Adulteration (Research) Aug 2024 – June 2025

UNDER THE GUIDANCE OF DR. MOHD MANSOOR KHAN, ASSISTANT PROFESSOR, IIIT GUWAHATI

- Engineered a low-cost, portable spectrophotometer using a compact spectroscopy sensor and a custom IoT platform.
- Developed a cross-platform (web and Android) application using React Native to provide real-time data visualization and analysis of turmeric samples.

Win32 System-Level Desktop Assistant & Telemetry Tool Mar 2024 – Dec 2024

- Engineered a stealth background process using C++ and the Win32 API, implementing global low-level hooks to intercept system message queues and simulate driver-level input handling.
- Manipulated OS process attributes and Window Handles (HWND) to decouple the execution instance from the taskbar and visual interface, demonstrating deep control over process visibility and memory management.

Experience & Leadership

Research Intern (SPARC Program) IIT Hyderabad Nov 2025

- Studied the end-to-end VLSI manufacturing process (Lithography, Etching, Deposition), bridging the gap between hardware design and physical silicon constraints.
- Analyzed device physics and scaling challenges, enhancing domain knowledge relevant to low-level firmware and hardware-software co-design.

Coordinator - Alankrit Fine-Arts Club – IIIT Guwahati Dec - Present

- Led 5+ campus-wide arts events, managing logistics, promotion, and volunteer teams; fostered artistic expression across campus as the club's most active coordinator.

Core Technical Member - E-Cell, TechnoCrats, Entropy Fest – IIIT Guwahati 2024 - 2025

- **(E-Cell)** Spearheaded the development of a full-stack prototype for the E-Cell website utilizing React and Node.js, completing the project in under 4 weeks, improving deployment efficiency by 20% and exceeding initial project expectations.

Achievements

Hackathon Finalist, India International Science Festival (IISF) 2024

- Recognized as a finalist team in the top 0.2% (from 45,000+ competing teams) in the national-level S&T Hackathon.

First Prize, GitHub University Blind Coding Challenge 2023

- Ranked #1 among all first-year students across every engineering discipline in a university-wide competition hosted by GitHub.